Credit Risk Intelligence Platform

An AI-powered platform for loan default prediction, explainability, and natural-language data access

NeoStats AI Engineer Internship — Candidate Assignment

OBJECTIVE

Building a bank-ready credit risk platform

Multi-table data

Joined application, bureau, prior-loan, and monthly repayment history — not just a single table

Explainable by design

Every prediction is backed by SHAP reasoning and auditable business rules, not a black box

Talk to the data

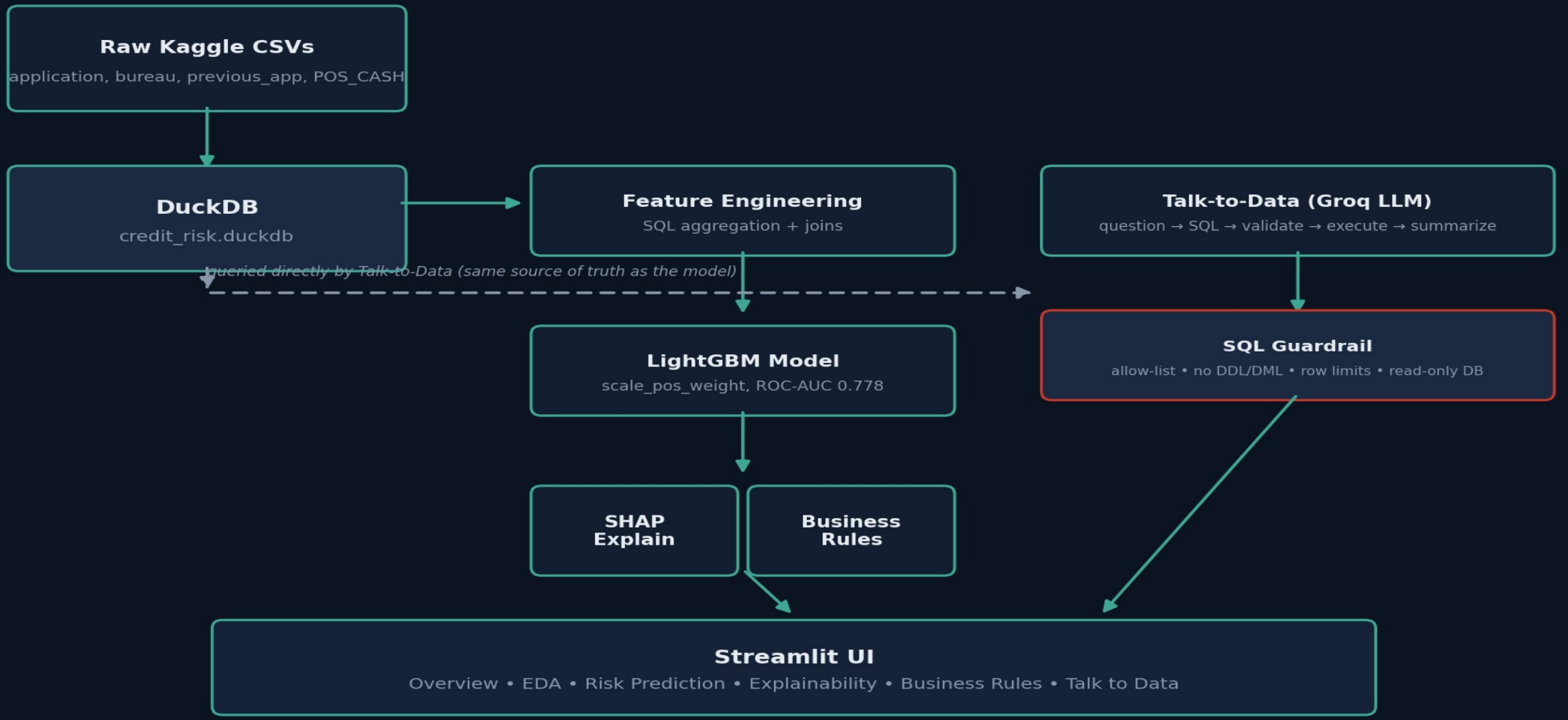
A guardrailed NL-to-SQL chatbot for non-technical stakeholders to explore the data directly

Ship it

Fully Dockerized, single-command deployment

ARCHITECTURE

How the pieces fit together



DATA

Multi-table feature engineering, not just one CSV

Most submissions on this dataset use application_train.csv alone. This platform joins three additional tables — aggregated in SQL for speed on 10M+ row tables — to capture signal a single-table model would miss.

applications

`307,511 rows`

Core applicant demographics & loan terms

bureau

`1.7M rows`

External credit bureau history — active/overdue loans, debt ratios

previous_applications

`1.7M rows`

This applicant's prior loans with the same lender

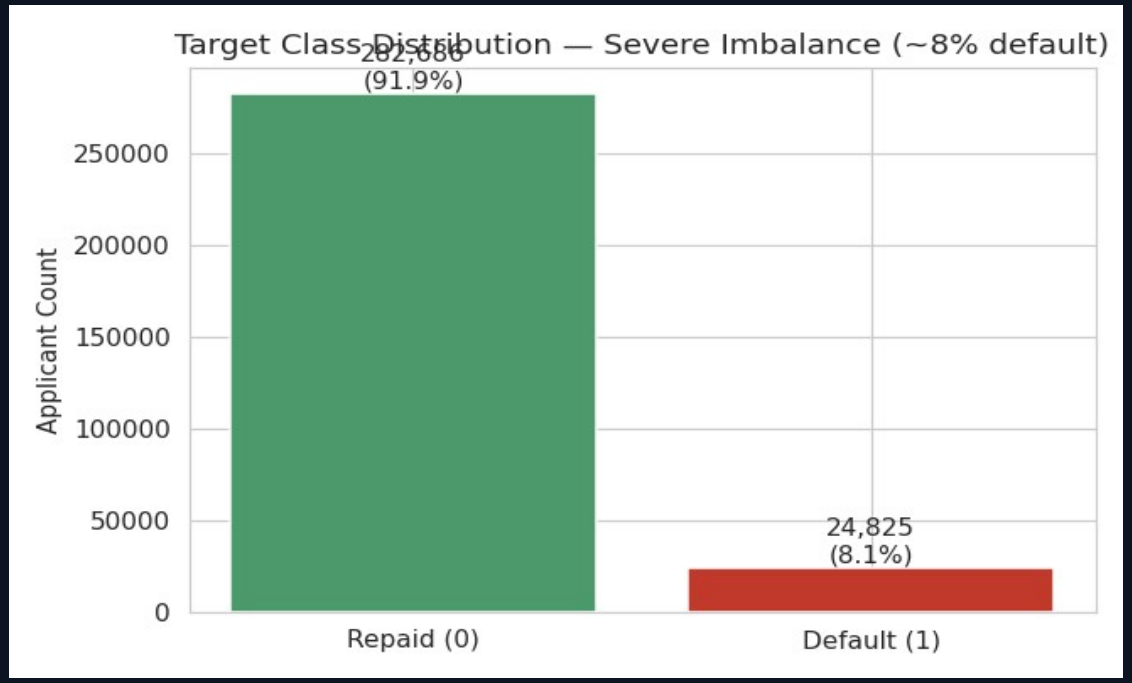
pos_cash_balance

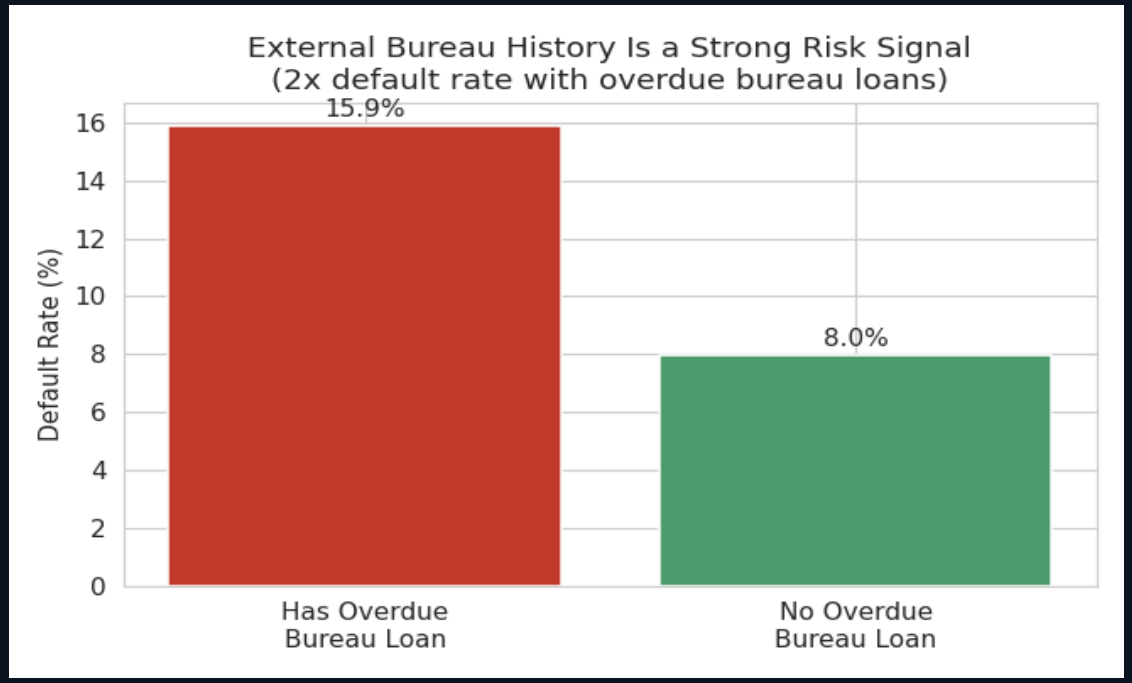
`10M rows`

Monthly days-past-due (DPD) — direct repayment behavior signal

Result: 152 features across 4 tables, feeding a single unified model.

What the data says before any model is built





- Only 8.07% of applicants default — a naive "approve everyone" model would be 92% "accurate" and useless
- An overdue external bureau loan nearly doubles default risk (15.9% vs 8.0%)

MACHINE LEARNING

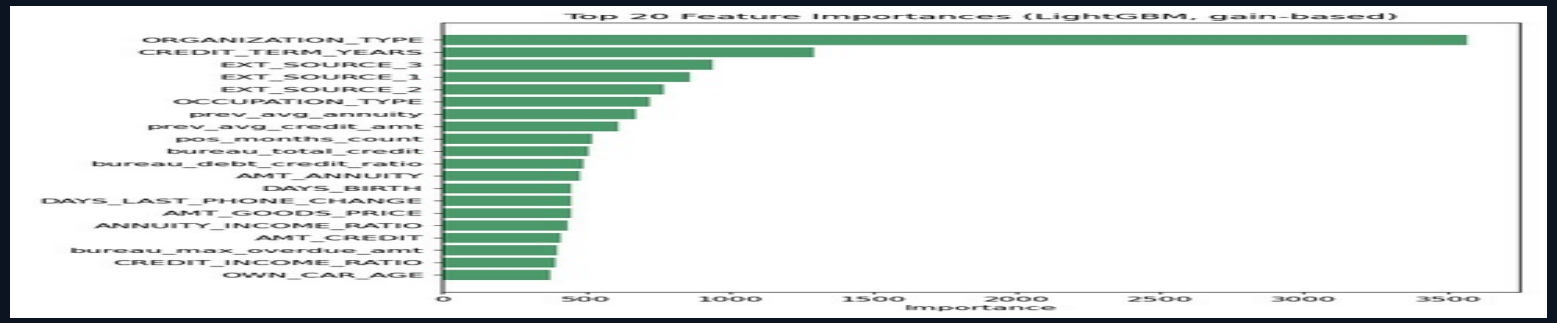
LightGBM with an evidence-based imbalance strategy

ROC-AUC	PR-AUC	Recall @ 0.5	Features
0.778	0.273	66.2%	152

Class imbalance (~8% default rate): scale_pos_weight vs SMOTE — tested, not assumed

Approach	Validation ROC-AUC	Notes
scale_pos_weight (production)	**0.7782**	Simpler, faster, no synthetic data
SMOTE (0.5 ratio)	0.7793	Statistically indistinguishable (difference = 0.001)

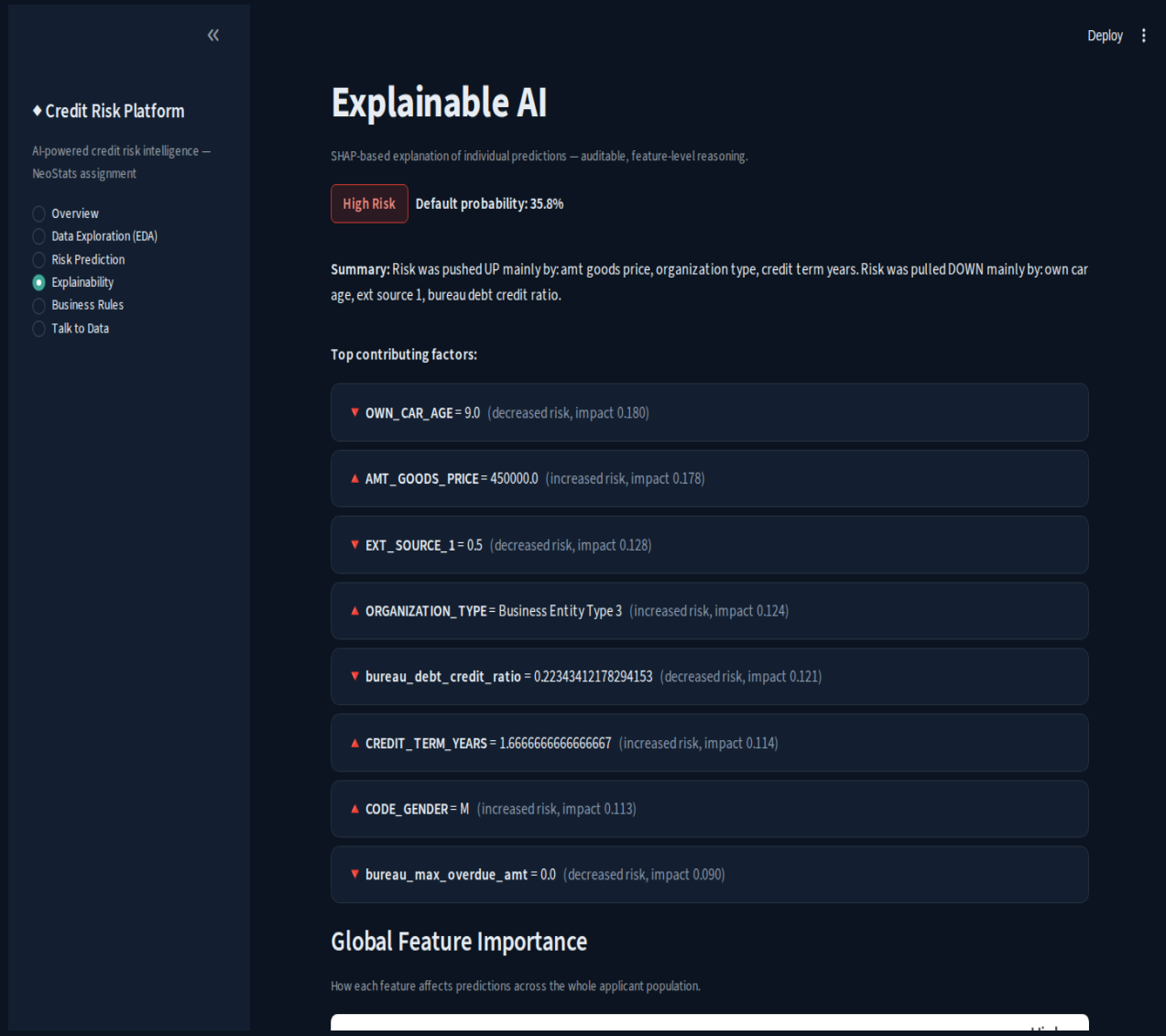
Evaluated on ROC-AUC / PR-AUC, not accuracy — accuracy is misleading under 8% class imbalance.



EXPLAINABLE AI

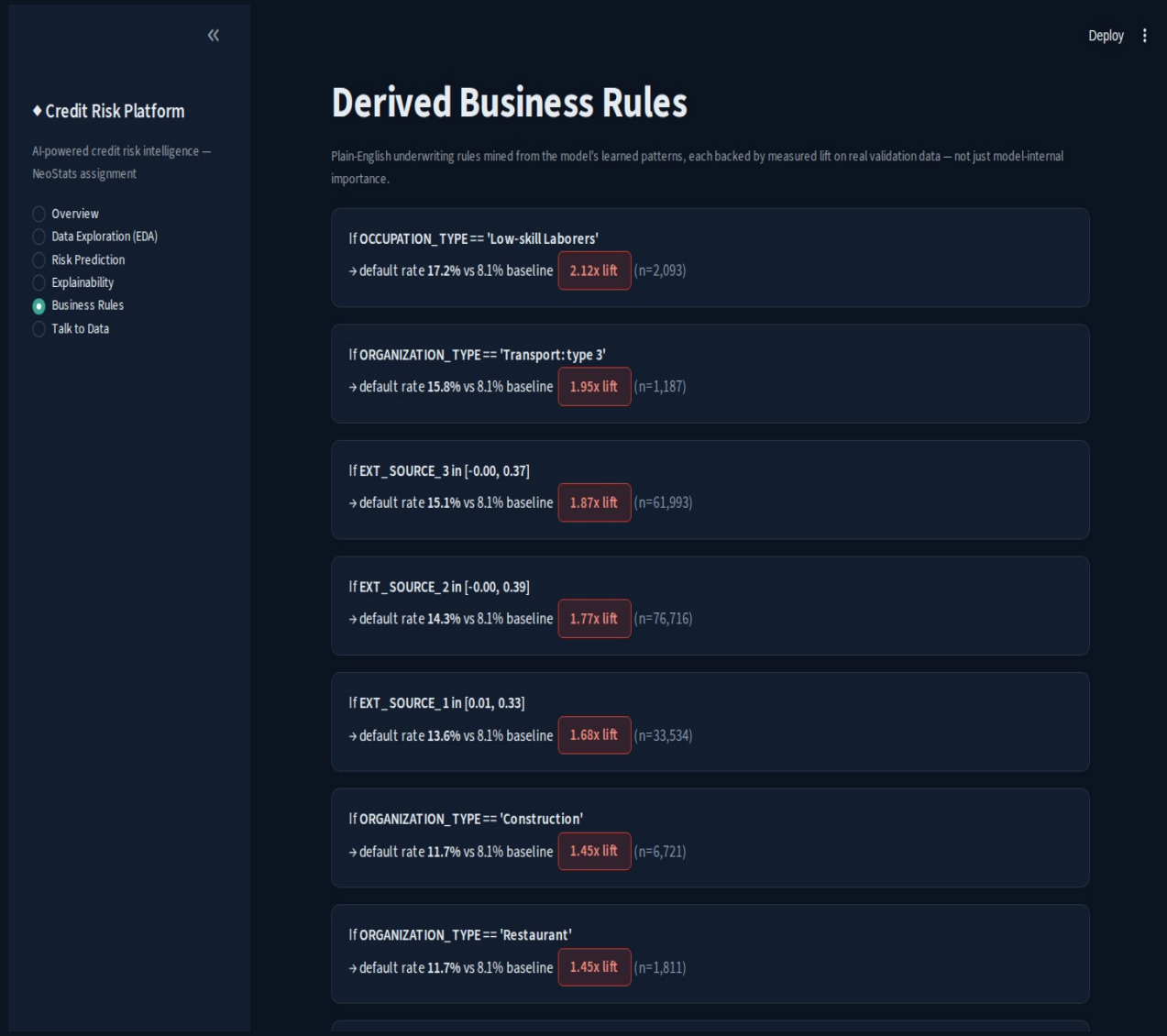
SHAP: auditable reasons behind every prediction

- TreeExplainer — exact SHAP values for gradient-boosted trees, not an approximation
- Per-applicant: top features, direction, and a plain-English summary
- Global: beeswarm plot across a 2,000-applicant sample
- Example: correctly flagged 22 active bureau loans + 70.7% refusal rate on a genuinely-defaulted applicant (p=0.955)



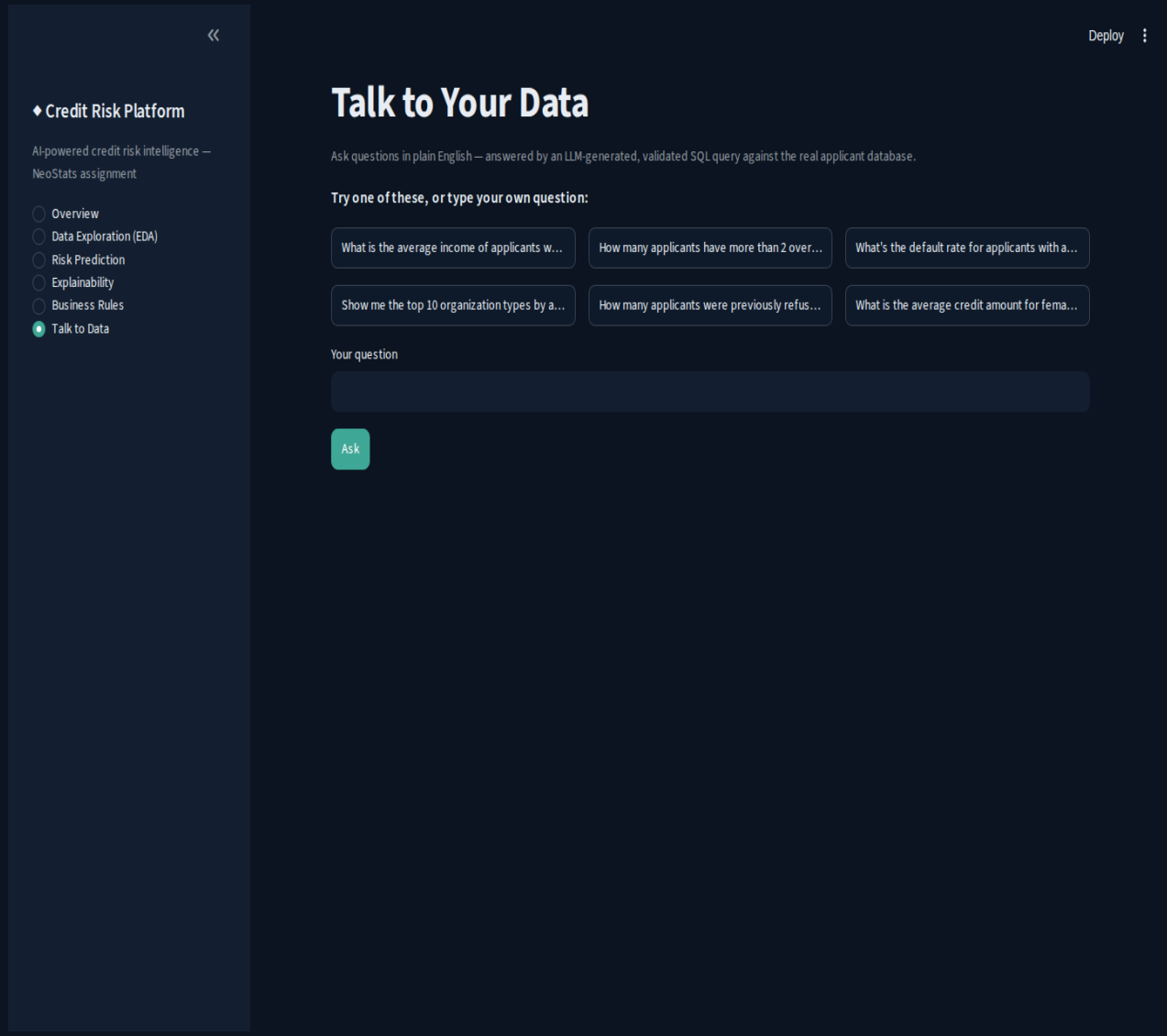
Bridging ML output and credit policy

- Bins top-25 model features and surfaces any bin where default rate is materially above baseline
- Every rule backed by measured lift + support (n), not just model-internal importance
- Auditable by a policy team independent of trusting the ML model
- Example: Low-skill Laborers default at 2.12x baseline (n=2,093)



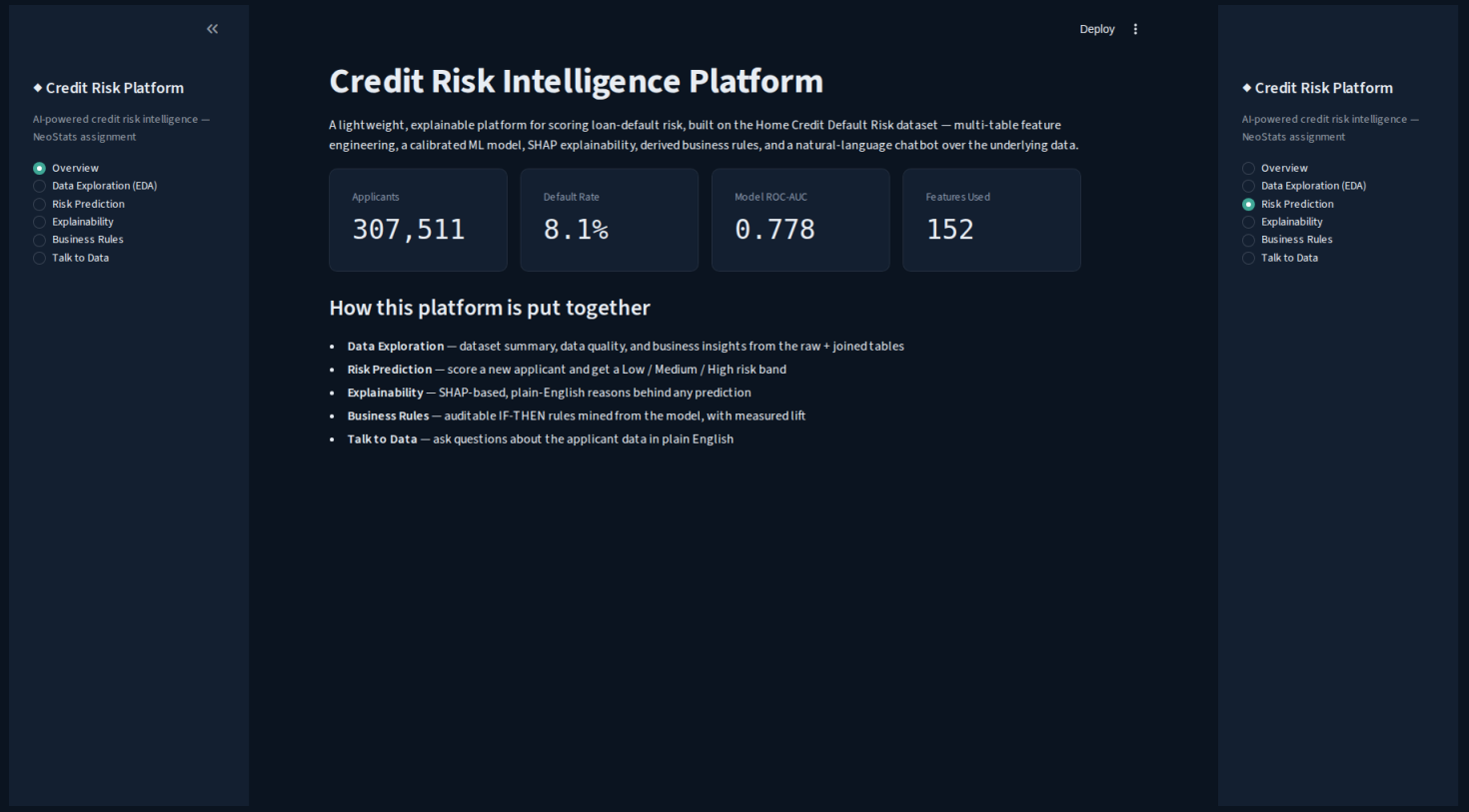
NL-to-SQL with guardrails enforced in code

- Two-stage LLM calls: SQL generation (temp 0.1) → grounded answer summarization from real returned rows
- SQL validator blocks: DDL/DML keywords, chained statements, hallucinated tables/columns
- Auto row-limiting + read-only DB connection as a second defense layer
- Tested against real adversarial inputs (DROP TABLE, injection, fake tables) — all correctly blocked
- 6 verified working query patterns (exceeds the required 5)

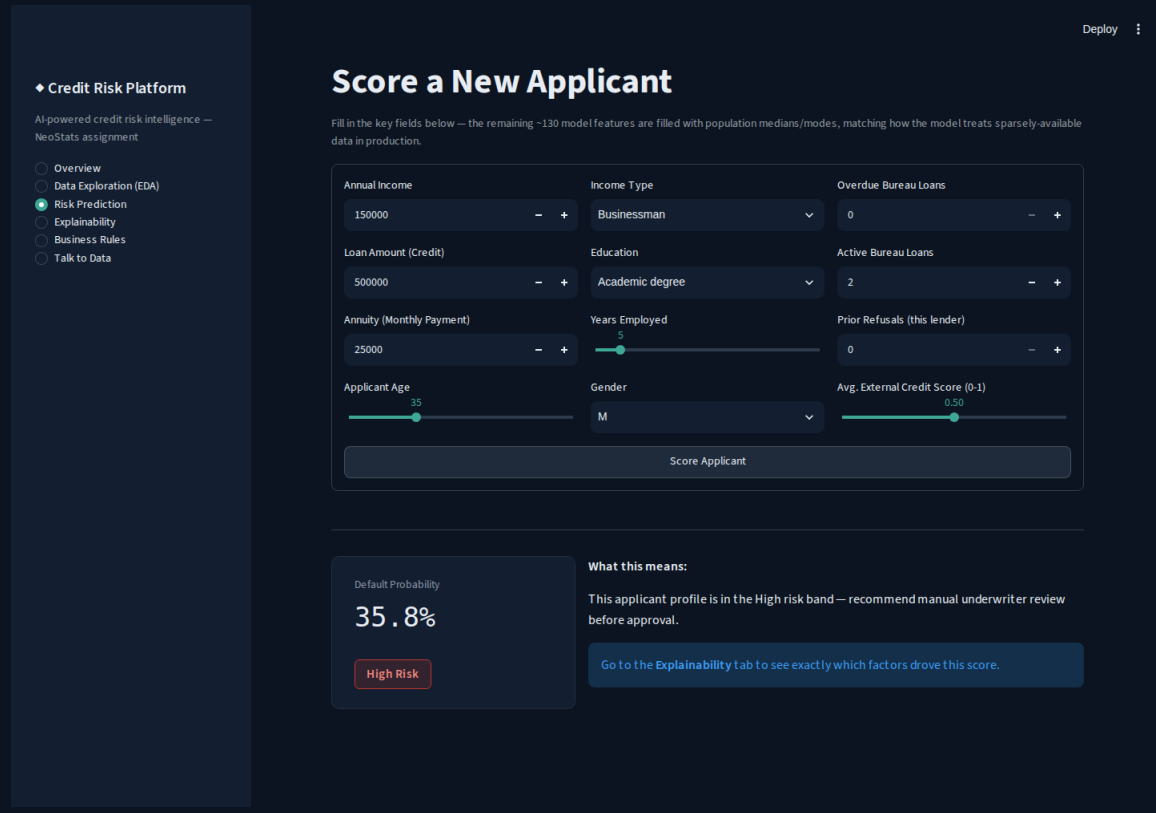


USER INTERFACE

One platform, five sections, real data throughout



Overview (KPIs from real data)



Risk Prediction (live scoring + risk band)

Streamlit • dark navy/teal design system • IBM Plex Sans • responsive KPI cards

DEPLOYMENT

One command to run the full platform

1

Clone & configure

git clone the repo, copy .env.example → .env, add a free Groq API key

2

Prep data & model

Download 5 Kaggle CSVs → run loader, preprocessor, train, evaluate, rules scripts

3

docker-compose up

Builds the image and launches Streamlit on port 8501 — single command

Dockerfile • docker-compose.yml • .env.example — all in the repo root, per the required structure

`github.com/malavika-krishna24/credit-risk-platform`

HONEST SCOPE

Known limitations & what's next

Two Kaggle tables not yet joined

bureau_balance.csv and credit_card_balance.csv would add further repayment signal on top of what's already used

Risk-band thresholds are defaults

Low/Medium/High cutoffs should be tuned against a real bank's approval/loss economics in production

Prediction form uses ~12 key fields

Remaining ~140 features filled from population medians/modes — a deliberate UX tradeoff for a demo interface

No monitoring/retraining pipeline

A production deployment needs drift detection and scheduled retraining — out of scope here

Thank you

Credit Risk Intelligence Platform — built end-to-end with real data, tested at every stage

github.com/malavika-krishna24/credit-risk-platform